

Difference Between Convert.ToString and ToString Method in C#

In this article, I am going to discuss the **Difference Between Convert.ToString and ToString Method in C#** with Examples. Please read our previous article before proceeding to this article where we discussed why and how to override the [Equals Method in C#](#) with examples.

Convert.ToString and ToString Method in C#

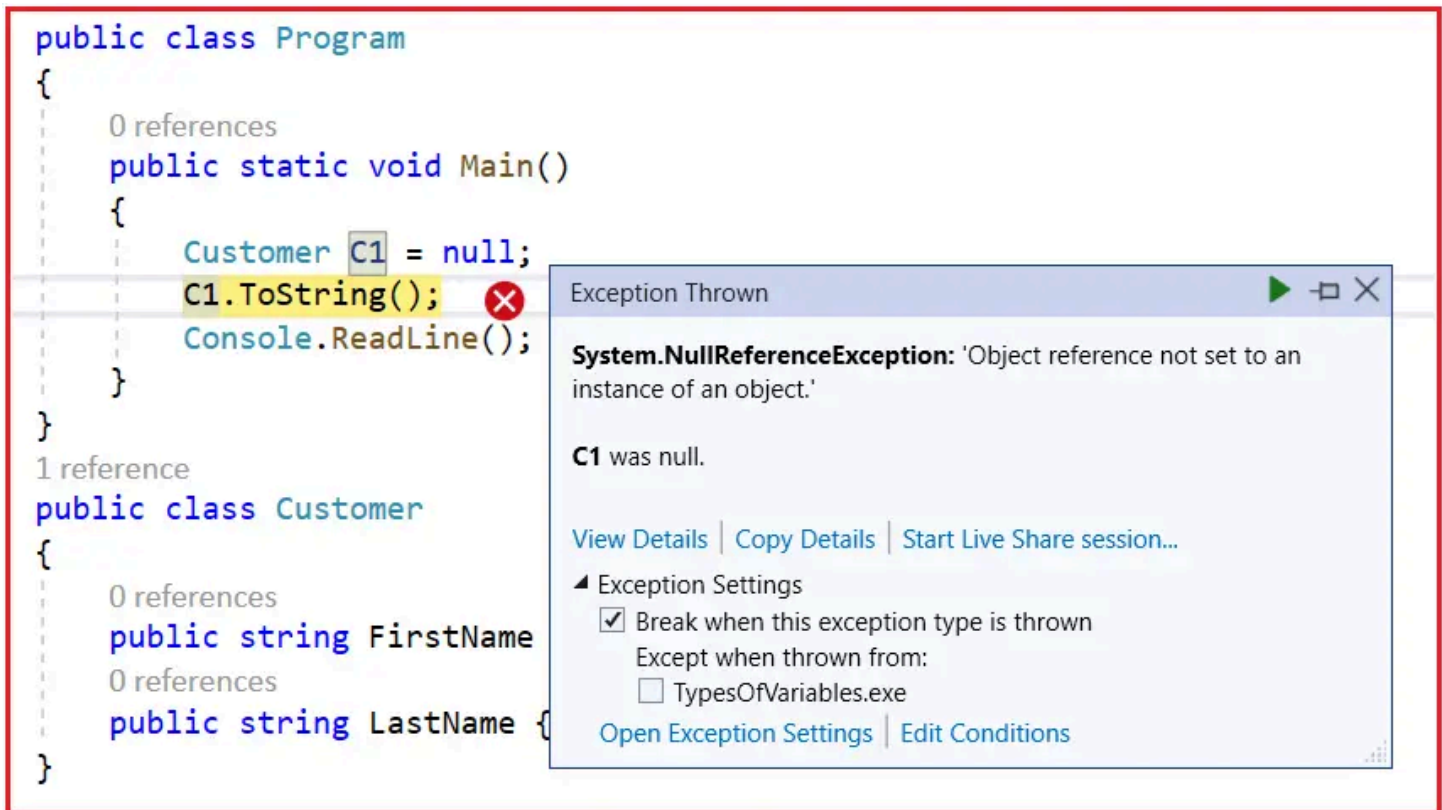
Both these methods are used to convert a value to a string. The difference is **Convert.ToString()** method handles null whereas the **ToString()** doesn't handle null in C#.

In C# if you declare a **string variable** and if you **don't assign any value** to that variable, then by default that variable takes a **null** value. In such a case, if you use the **ToString()** method then your program will throw the **Null Reference Exception**. On the other hand, if you use the **Convert.ToString()** method then your program will not throw an exception.

Let us understand the Difference Between these two methods with an example.

```
using System;
namespace UnderstandingToStringMethod
{
    public class Program
    {
        public static void Main()
        {
            Customer C1 = null;
            C1.ToString();
            Console.ReadLine();
        }
    }
    public class Customer
    {
        public string FirstName { get; set; }
        public string LastName { get; set; }
    }
}
```

When you run the application, it will give you the following error

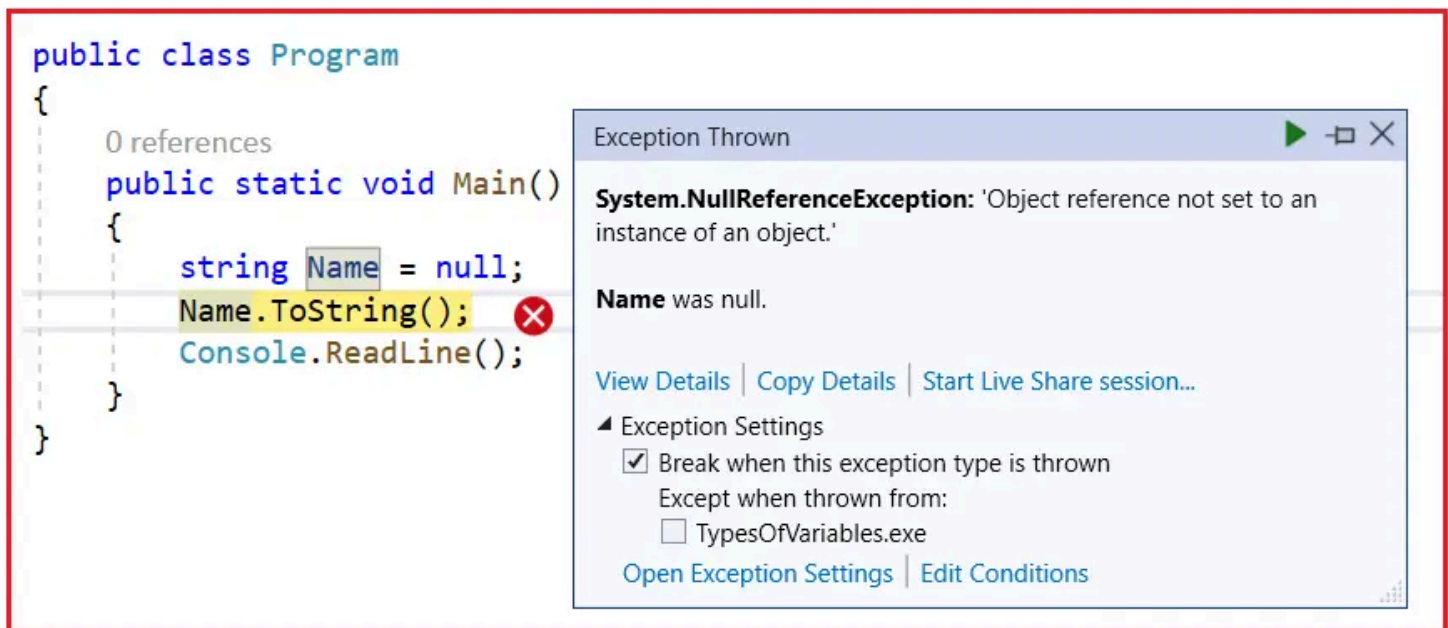


This is because the **ToString()** method in C# expects the object to be not NULL on which it is invoking. In our example object, **C1 is Null** and we are invoking the **ToString()** on the **NULL object**, so it gives **NULL Reference exception**.

Let see another example.

```
using System;
namespace UnderstandingToStringMethod
{
    public class Program
    {
        public static void Main()
        {
            String Name = null;
            Name.ToString();
            Console.ReadLine();
        }
    }
}
```

When we execute the above program, it also gives us the same **NULL Reference Exception** as shown below.



This is because the Name variable is Null and we are invoking the ToString() method. Let see what happens when we use the **Convert.ToString()** method with the above two examples.

```
using System;
namespace UnderstandingObjectClassMethods
{
    public class Program
    {
        public static void Main()
        {
            Customer C1 = null;
            Convert.ToString(C1);
            String Name = null;
            Convert.ToString(Name);
            Console.WriteLine("No Error");
            Console.ReadLine();
        }
    }
    public class Customer
    {
        public string FirstName { get; set; }
        public string LastName { get; set; }
    }
}
```

Now, with the above changes, run the application and it should be executed without any error. So in short, the **Convert.ToString()** method handles null, while the **ToString()** method doesn't handle the Null and throws an exception. So it's always a good programming practice to use the **Convert.ToString()** method which will take care of the Null values and it is also safe to use.

In the next article onwards, I am going to discuss **checked and unchecked keywords in C#** with Examples. Here, in this article, I try to explain the **difference between the Convert.ToString and ToString Method in C#** with Examples. I hope this article will help you with your needs. I would like to

have your feedback. Please post your feedback, question, or comments about this Difference Between Convert.ToString and ToString Method in C# article.